# Nader Khalil

## AI Engineer · Agentic Systems & Automation

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### PROFESSIONAL SUMMARY

AI Engineer with a B.Sc. in Software Engineering and Unix systems training from 42 Abu Dhabi, designing and deploying production Generative AI applications, RAG pipelines, and agent orchestration workflows. Proficient in vector search (pgvector, Pinecone/FAISS), prompt engineering, automated AI evaluation suites (prompt injection defense, citation grounding), and inference optimization (93% prompt caching). Experienced in mentoring UAE youth and engineering teams on agentic AI for ADEK. Bilingual in Arabic (Native) and English (Professional), based in Abu Dhabi.

### TECHNICAL SKILLS

**Generative AI & LLMs:** RAG architectures, Multi-Agent Orchestration, Tool Calling, MCP, n8n Automation, Prompt Caching, LangChain, Vercel AI SDK, GPT, Llama

**AI Safety & Evals:** Automated prompt injection testing, scope gating, structured output parsing (Zod, Pydantic), hallucination reduction, citation grounding

**MLOps, Cloud & CI/CD:** Docker, GitHub Actions CI/CD, Azure AI / Cloud platforms, automated test suites (766+ tests in CI), Playwright, Vitest, PyTest

**Vector DBs & Retrieval:** Vector embeddings, semantic search, pgvector, Pinecone, FAISS, PostgreSQL, MariaDB, SQL schema design, knowledge base ingestion

**Software & API Engineering:** RESTful APIs, Python (FastAPI), TypeScript, Next.js, React 19, layered architecture, Auth.js, RBAC, web service integration

**Languages & Core:** Python, TypeScript, JavaScript, SQL, C, C++, Bash, Linux system internals

### TECHNICAL PROJECTS

**AI Booking Platform,** *Production GenAI & Agentic System* 2026
Live build: booking.naderkhalil.dev · Case study: naderkhalil.dev/case-study.html

- Architected an end-to-end RAG pipeline querying admin-managed docs with vector embeddings and role-filtered SQL retrieval (pgvector); answers include verifiable source citations and enforce conversational context isolation.
- Designed agent orchestration with autonomous tool calling and Human-in-the-Loop review: the model proposes calendar actions as structured cards executed only upon explicit user confirmation.
- Built an automated evaluation suite with 120 live checks gating prompt injections, out-of-scope queries, and data leaks; achieved 93% cached input inference efficiency and verified concurrency with 766 tests in CI.

**AI Appointment Scheduler,** *LangChain Conversational Agent* 2025
Live build: appointment.naderkhalil.dev

- Built a conversational scheduling agent using Python (FastAPI), LangChain, and Next.js, parsing natural-language scheduling requests into structured API payloads for calendar availability checks and slot reservations.

**Student-Faculty Services v2,** *Higher-Ed Service Platform* 2026
Live build: student-faculty-services-v2.infinityfree.io · Case study: naderkhalil.dev/case-study-v2.html

- Architected a high-reliability educational services API (46 endpoints) in PHP 8.4 and React 19; eliminated double-bookings via database row locks (proven in 8-student race tests); verified by 500+ tests with 100/100 accessibility.

### PROFESSIONAL EXPERIENCE

**AI Trainer & Mentor,** *Department of Education and Knowledge (ADEK)* 07/2026 – 08/2026

- Delivered technical instruction on agentic workflows, multi-agent orchestration (AG2), and autonomous tool calling in n8n for the accredited AgenticVerse program; mentored teams building AI prototypes for ADNOC, FANR, and Khalifa Fund.
- Guided participants through API schema validation, RAG grounding, safety gates (prompt injection defense, human review), and debugging in Python and TypeScript.

**QA & Software Tester,** *42 Abu Dhabi* 01/2026

- Conducted functional and UI testing on internal academic platforms using black-box scenarios; authored reproduction reports with logs and steps to help developers isolate defects quickly.
- Tested features across browsers and viewports; re-tested defect fixes across user flows to ensure zero regressions before release.

**IT Support & Technical Staff,** *Ticketmaster* 12/2025

- Provisioned and deployed entry ticketing hardware and access scanners for a high-volume international event; maintained 100% operational uptime through real-time hardware health checks and rapid on-site incident resolution.

**Software Engineering Intern,** *Timex IT Consulting* 01/2024 – 04/2024

- Developed C# MinimalAPI CRUD endpoints with request validation and standardized HTTP response handling; verified functional defect fixes on client accounting software before production release.

### EDUCATION & CERTIFICATIONS

**B.Sc. in Software Engineering,** *Al Ain University*
**Computer Science Diploma,** *42 Abu Dhabi*
**MOEI × 42 Abu Dhabi AgentEra Hackathon,** *Honoree*

**CyberQ CTF, Top 10 Finalist,** *UAE Cybersecurity Council*
**Hack The Box University CTF,** *Top 12% (119/1,014 teams)*
**RecognAIze Hackathon,** *AI on Watch with Datamind*

**Languages:** Arabic (Native), English (Professional Proficiency)